assignment1

January 20, 2025

1 Objectives

The objective of this assignment is to apply the concepts and techniques of Exploratory Data Analysis (EDA) to a real-world dataset about adult income in the USA during the 1990s. By leveraging the skills learned in Chapter 2 (EDA) and Labs 1 and 2, this assignment aims to explore the dataset, which contains various demographic and employment-related variables. The key focus is on analyzing both categorical and quantitative variables, creating visualizations, and calculating summary statistics to draw meaningful insights from the data.

This assignment will involve the following tasks: 1. **Exploring categorical variables**: Analyzing variables like workclass, education, marital status, occupation, sex, and race. 2. **Exploring quantitative variables**: Analyzing variables like age, education number, and hours worked per week. 3. **Performing visual analysis**: Creating frequency distribution tables, pie charts, bar charts, histograms, and box plots to visualize the distribution of variables. 4. **Calculating statistical measures**: Computing key statistics such as mean, median, mode, standard deviation, range, interquartile range (IQR), and identifying outliers.

The ultimate goal of this assignment is to understand the structure of the dataset, identify patterns, and summarize the key characteristics of the data. The insights gained through this analysis will provide a solid foundation for further statistical analysis or predictive modeling.

1.0.1 Loading the Dataset:

To begin analyzing the dataset, the following Python code can be used to read the CSV file into a pandas DataFrame:

[235]:
```python
import pandas as pd

# Path to the cleaned census income dataset
file_path = 'data/cleaned_census_income.csv'

# Read the dataset into a pandas DataFrame
df = pd.read_csv(file_path)

# Display the first few rows of the dataset
df
```

[235]:
```
        age workclass  fnlwgt     education  education.num      marital.status  \
0        82   Private  132870       HS-grad              9            Widowed
```

```
1        54   Private  140359      7th-8th              4           Divorced
2        41   Private  264663  Some-college             10          Separated
3        34   Private  216864       HS-grad              9           Divorced
4        38   Private  150601          10th              6          Separated
…     …          …      …            …            …               …
30157    22   Private  310152  Some-college             10      Never-married
30158    27   Private  257302    Assoc-acdm             12  Married-civ-spouse
30159    40   Private  154374       HS-grad              9  Married-civ-spouse
30160    58   Private  151910       HS-grad              9            Widowed
30161    22   Private  201490       HS-grad              9      Never-married

             occupation   relationship   race     sex  capital.gain  \
0       Exec-managerial  Not-in-family  White  Female             0
1     Machine-op-inspct      Unmarried  White  Female             0
2        Prof-specialty      Own-child  White  Female             0
3         Other-service      Unmarried  White  Female             0
4          Adm-clerical      Unmarried  White    Male             0
…                  …              …     …       …            …
30157   Protective-serv  Not-in-family  White    Male             0
30158      Tech-support           Wife  White  Female             0
30159 Machine-op-inspct        Husband  White    Male             0
30160      Adm-clerical      Unmarried  White  Female             0
30161      Adm-clerical      Own-child  White    Male             0

      capital.loss  hours.per.week native.country income
0             4356              18  United-States  <=50K
1             3900              40  United-States  <=50K
2             3900              40  United-States  <=50K
3             3770              45  United-States  <=50K
4             3770              40  United-States  <=50K
…             …             …             …     …
30157            0              40  United-States  <=50K
30158            0              38  United-States  <=50K
30159            0              40  United-States   >50K
30160            0              40  United-States  <=50K
30161            0              20  United-States  <=50K

[30162 rows x 15 columns]
```

1.0.2 Q1) What are the categorical variables in this dataset, and why?

Categorical Variables: Categorical variables are those that represent categories or groups. These variables can take on a limited number of distinct values and usually do not have any inherent numerical meaning, though some may have an ordinal relationship (i.e., an order to the categories). In this dataset, the following are categorical variables:

1. **Workclass**:
    - Represents the type of employment or work status of an individual. Examples include

    "Private", "Self-emp-not-inc", "State-gov", etc. This variable is categorical because it represents categories of employment status.

2. **Education**:
    - Indicates the highest level of education attained by the individual (e.g., "HS-grad", "Bachelors", "Doctorate", etc.). This is categorical because it groups individuals based on their educational achievements.
3. **Marital.status**:
    - Describes the individual's marital status, such as "Married", "Divorced", "Widowed", "Never married", etc. This is categorical because it denotes the relationship status of the individual.
4. **Occupation**:
    - Represents the type of occupation the individual holds (e.g., "Exec-managerial", "Prof-specialty", "Machine-op-inspct", etc.). Since occupations are distinct categories, this is a categorical variable.
5. **Relationship**:
    - Represents the individual's relationship to others in the household, such as "Husband", "Wife", "Own child", "Not in family", etc. This is categorical because it categorizes individuals based on familial relationships.
6. **Race**:
    - Represents the race of the individual, such as "White", "Black", "Asian Pac Islander", etc. This is a categorical variable because it consists of distinct racial categories.
7. **Sex**:
    - The biological sex of the individual, with categories like "Male" and "Female". This is categorical because it divides the population into two groups based on biological sex.
8. **Native.country**:
    - Indicates the individual's country of origin (e.g., "United-States", "Mexico", "India", etc.). This is a categorical variable because it represents distinct countries or origins.
9. **Income**:
    - Represents the income level of the individual, with categories like ">50K" and "<=50K". This is categorical because it categorizes individuals based on their income level.

Why These Variables Are Categorical:

- These variables are considered categorical because their values represent distinct categories or groups without inherent numerical meaning (with the exception of **education.num**, which is a numerical encoding of education levels).
- For example, "Workclass" categorizes individuals into different employment statuses, but the categories themselves don't imply a ranking or order (i.e., no inherent numerical relationships exist between "Private" and "Self-emp-not-inc").
- **Race**, **Sex**, **Relationship**, and **Marital.status** also fit the same pattern—they divide the population into non-ordinal categories.

1.0.3 Summary:

The dataset contains several categorical variables such as workclass, education, marital status, occupation, race, sex, native country, and income. These variables group individuals into distinct categories, which are crucial for analyzing demographic patterns and conducting further statistical analysis.

1.0.4 Q2) What are the quantitative variables in this dataset, and why?

Quantitative Variables: Quantitative variables are those that are numerical in nature and can be measured on a numerical scale. These variables represent amounts, quantities, or counts, and they can be subjected to mathematical operations such as addition, subtraction, multiplication, and division. The quantitative variables in this dataset are as follows:

1. **Age**:
   - Represents the age of an individual, measured in years. This is a quantitative variable because age is a numerical value that can be measured and used in arithmetic operations.
2. **fnlwgt**:
   - The final weight, representing the number of people that the entry is assumed to represent. This is a quantitative variable because it is a numerical value that reflects the weight or importance of a data entry in the census dataset.
3. **Education.num**:
   - This is the numerical representation of the highest level of education achieved by the individual. For example, "HS-grad" might be represented as 9, and "Doctorate" as 16. This variable is quantitative because it involves numerical values that can be used in mathematical operations, even though it's a coded representation of educational levels.
4. **Capital.gain**:
   - Represents the amount of capital gain earned by an individual, which is a numerical value (e.g., 0, 4356, etc.). This is a quantitative variable because it is a measurable amount of income from capital investments.
5. **Capital.loss**:
   - Represents the amount of capital loss incurred by an individual, also a numerical value (e.g., 0, 4356, etc.). Similar to capital gain, this is quantitative as it represents measurable losses from capital investments.
6. **Hours.per.week**:
   - Represents the number of hours an individual works per week, measured in hours. This is a quantitative variable because it is a numerical value that can be used to quantify time worked and can be mathematically manipulated.

Why These Variables Are Quantitative:

- These variables are considered quantitative because they represent measurable quantities or amounts, and they are expressed in numerical form. Unlike categorical variables, which represent categories, quantitative variables allow for meaningful mathematical operations, such as calculating averages, sums, and standard deviations.
- For example, **Age** can be summed, averaged, and used to calculate differences, while **Capital.gain** and **Capital.loss** represent financial amounts that can be mathematically analyzed.
- **Education.num** is a coded variable representing different education levels but still retains a numerical meaning, allowing it to be treated as a quantitative variable for analysis.

Summary: The dataset contains several quantitative variables, including age, fnlwgt, education.num, capital gain, capital loss, and hours worked per week. These variables are numerical and allow for various mathematical and statistical analyses, such as calculating averages, variances, and other summary statistics. Understanding these quantitative variables is crucial for analyzing the distribution of income and other demographic patterns in the dataset.

1.0.5 Q3) Examine Distribution of Workclass Variable

To examine the distribution of the **workclass** variable, we will follow these steps:

1. **Create a frequency distribution table** for the different values of the **workclass** variable.
2. **Plot a pie chart** to visually represent the distribution of workclass categories.
3. **Plot a bar chart** to visualize the frequency of each workclass category.
4. **Provide a summary/interpretation** of the distribution of the **workclass** variable.

Step 1: Frequency Distribution Table for Workclass Variable We begin by creating a frequency distribution table that shows the count of each unique category in the **workclass** variable. This will give us an overview of how many individuals belong to each employment status category.

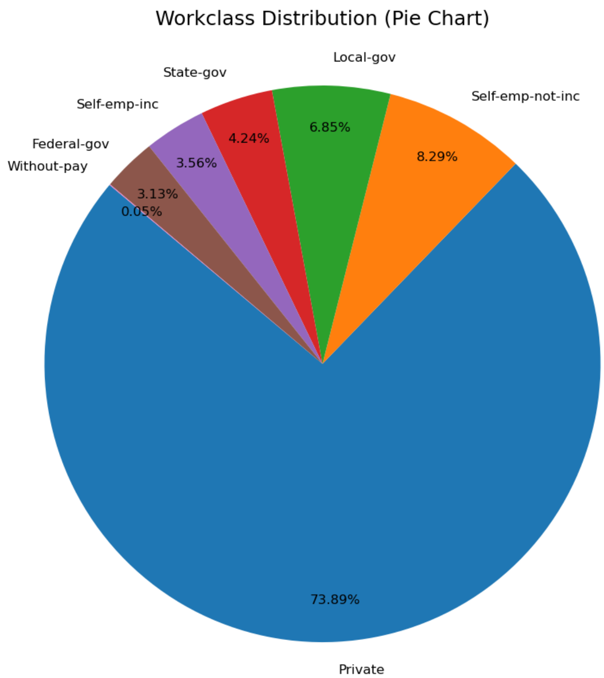
[241]:
```
# Step 1: Frequency Distribution Table for Workclass
workclass_counts = df['workclass'].value_counts()

# Display the frequency distribution table
print("Frequency Distribution Table for Workclass:")
print(workclass_counts)
```

```
Frequency Distribution Table for Workclass:
workclass
Private             22286
Self-emp-not-inc     2499
Local-gov            2067
State-gov            1279
Self-emp-inc         1074
Federal-gov           943
Without-pay            14
Name: count, dtype: int64
```

Step 2: Pie Chart for Workclass Variable Next, we plot a pie chart to visually represent the proportion of each category in the **workclass** variable. This helps us quickly understand the distribution and relative sizes of the categories.

[244]:
```
import matplotlib.pyplot as plt

# Step 2: Pie Chart for Workclass Variable with 2 decimal places
plt.figure(figsize=(18, 10))  # Larger chart size for better readability
plt.pie(workclass_counts,
        labels=workclass_counts.index,
        autopct='%1.2f%%',  # Showing percentage to 2 decimal places
        startangle=140,
        pctdistance=0.85,  # Adjusts label distance from the center
        labeldistance=1.1,  # Increased space between labels and chart
        textprops={'fontsize': 12})  # Slightly adjusted font size for␣
 ↪readability
```

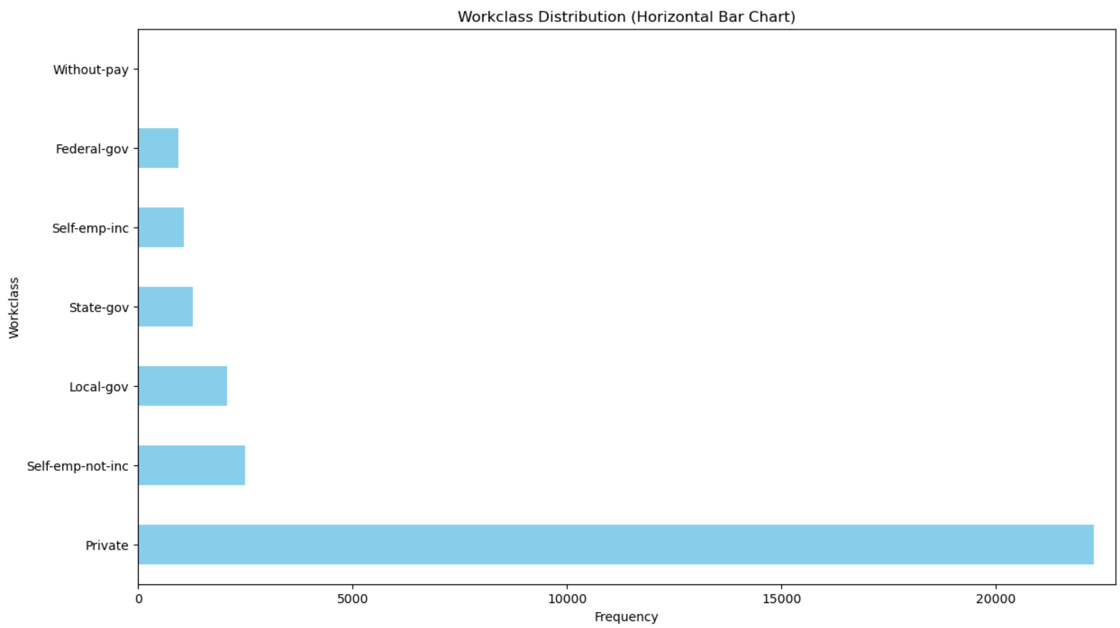
```
# Add a title to the chart with some padding
plt.title('Workclass Distribution (Pie Chart)', fontsize=18, pad=30)

# Ensure that the pie chart is circular
plt.axis('equal')

# Display the pie chart
plt.show()
```



Step 3: Bar Chart for Workclass Variable We will now plot a bar chart to visualize the frequency of each **workclass** category. A bar chart provides a clearer view of the count of individuals in each category, allowing for easy comparison between categories.

[247]:
```
# Step 3: Horizontal Bar Chart for Workclass Variable with Adjustments
plt.figure(figsize=(14, 8))  # Larger figure size for better visibility of all␣
↪categories
workclass_counts.plot(kind='barh', color='skyblue')  # Horizontal bar chart

# Format the x-axis and labels
plt.title('Workclass Distribution (Horizontal Bar Chart)')
plt.xlabel('Frequency')
plt.ylabel('Workclass')

# Adjust x-axis limit to ensure visibility of all categories (including␣
↪"Without-pay")
```

```
plt.xlim(0, workclass_counts.max() + 500)  # Adding more space to the right for
 ↪small categories

# Display the chart
plt.show()
```



Step 4: Interpretation of the Workclass Distribution

Frequency Distribution Table: The frequency distribution table for the **workclass** variable is as follows:

- **Private**: 22,286 (73.9%)
- **Self-emp-not-inc**: 2,499 (8.3%)
- **Local-gov**: 2,067 (6.9%)
- **State-gov**: 1,279 (4.2%)
- **Self-emp-inc**: 1,074 (3.6%)
- **Federal-gov**: 943 (3.1%)
- **Without-pay**: 14 (0.05%)

Pie Chart Distribution: The pie chart clearly shows the proportion of individuals in each **workclass** category:

- **Private** employment is the most prevalent, comprising **73.9%** of the dataset. This suggests that most individuals in this dataset work in private-sector jobs.
- **Self-emp-not-inc** follows with **8.3%**, representing individuals who are self-employed but not incorporated.

- **Local-gov** and **State-gov** contribute **6.9%** and **4.2%**, respectively, representing those working in local and state government sectors.
- **Self-emp-inc** accounts for **3.6%**, and **Federal-gov** makes up **3.1%**, showing the proportion of individuals employed in incorporated self-employment and federal government sectors.
- The **Without-pay** category, although present, is very small, representing only **0.05%** of the dataset. In the pie chart, it's visible but not dominant, highlighting the small proportion of individuals working without pay.

Bar Chart Distribution: While the **pie chart** clearly represents the **Without-pay** category, the **horizontal bar chart** does not show this category effectively, even after adjusting the figure size and axis limits. The **Without-pay** category, with only **14 individuals**, is too small to be visually distinguished in the bar chart.

Despite the large figure size and axis adjustment, **Without-pay** is still not visible in the bar chart because the rest of the categories dominate the frequency scale. This is expected, given the extremely low count for this category.

Summary:

- The **Private** sector overwhelmingly dominates the dataset, comprising more than **73%** of individuals.
- **Self-emp-not-inc** and other self-employment categories account for **approximately 12%** of the dataset.
- The **Without-pay** category, representing a mere **0.05%** of the dataset, is too small to be visible in the bar chart, even with adjustments to the axis and figure size. This highlights the very small number of individuals working without pay in this dataset.
- The **pie chart** is a better tool for showing smaller categories like **Without-pay**, while the **bar chart** provides a clear comparison for the more substantial workclass categories.

1.0.6 Q4) Examine Distribution of Education Variable

To examine the distribution of the **education** variable, we will follow these steps:

1. **Create a frequency distribution table** for the different values of the **education** variable.
2. **Plot a pie chart** to visually represent the distribution of education categories.
3. **Plot a bar chart** to visualize the frequency of each education category.
4. **Provide a summary/interpretation** of the distribution of the **education** variable.

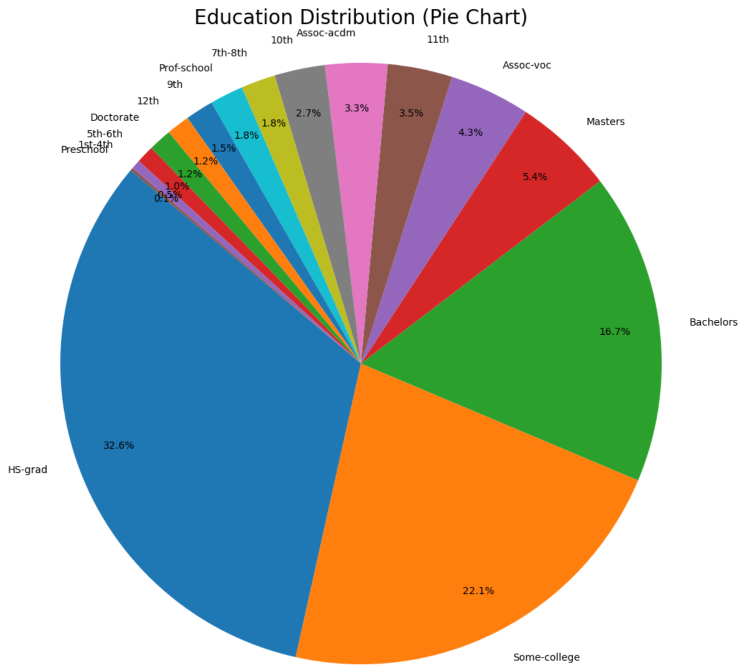
Step 1: Frequency Distribution Table for Education Variable First, we'll create a frequency distribution table that shows the count of each unique category in the **education** variable.

[252]:
```
# Step 1: Frequency Distribution Table for Education
education_counts = df['education'].value_counts()

# Display the frequency distribution table
print("Frequency Distribution Table for Education:")
print(education_counts)
```

```
Frequency Distribution Table for Education:
education
HS-grad         9840
Some-college    6678
Bachelors       5044
Masters         1627
Assoc-voc       1307
11th            1048
Assoc-acdm      1008
10th             820
7th-8th          557
Prof-school      542
9th              455
12th             377
Doctorate        375
5th-6th          288
1st-4th          151
Preschool         45
Name: count, dtype: int64
```

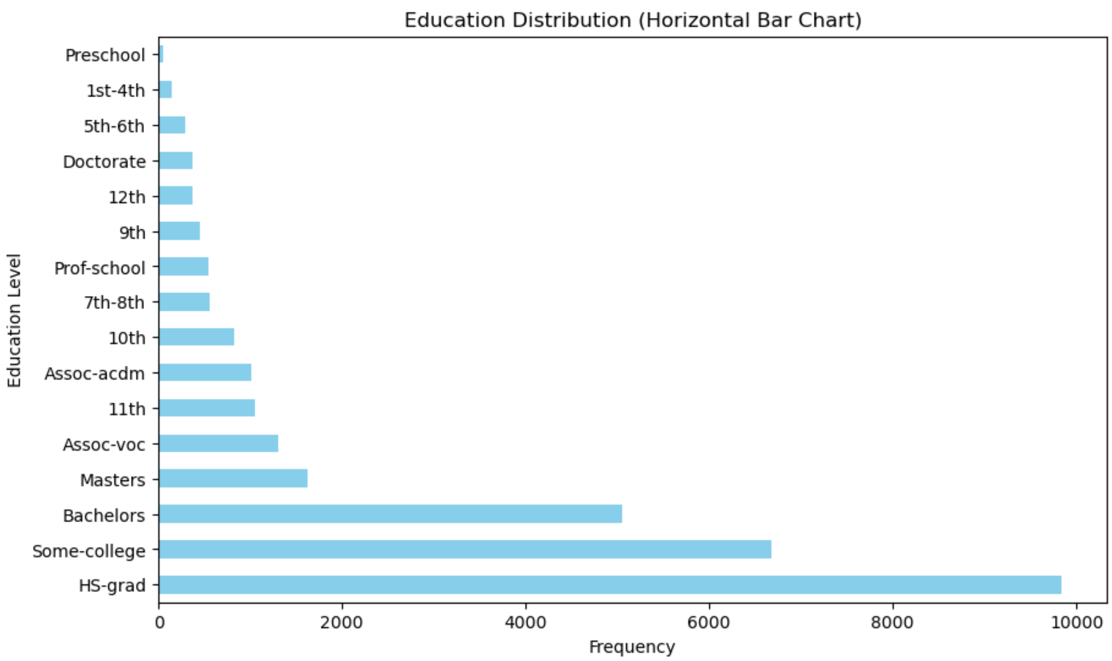
Step 2: Pie Chart for Education Variable Next, we'll plot a pie chart to visually represent the proportion of each category in the **education** variable.

[255]:
```python
# Step 2: Pie Chart for Education Variable
plt.figure(figsize=(20, 12))  # Significantly large figure size
plt.pie(education_counts,
        labels=education_counts.index,
        autopct='%1.1f%%',  # Showing percentage to 1 decimal place
        startangle=140,
        pctdistance=0.85)  # Adjusts the label distance from the center

# Add a title to the chart
plt.title('Education Distribution (Pie Chart)', fontsize=20, pad=10)

# Ensure that the pie chart is circular
plt.axis('equal')

# Display the pie chart
plt.show()
```



Step 3: Bar Chart for Education Variable Now, we will plot a bar chart to visualize the frequency of each **education** category.

[258]:
```python
# Step 3: Bar Chart for Education Variable
plt.figure(figsize=(10, 6))
education_counts.plot(kind='barh', color='skyblue')  # Horizontal bar chart

# Format the x-axis and labels
plt.title('Education Distribution (Horizontal Bar Chart)')
plt.xlabel('Frequency')
plt.ylabel('Education Level')

# Display the chart
plt.show()
```



Step 4: Interpretation of the Education Distribution

Frequency Distribution Table: The frequency distribution table for the **education** variable is as follows:

- **HS-grad**: 9,840 (32.6%)
- **Some-college**: 6,678 (22.1%)
- **Bachelors**: 5,044 (16.7%)
- **Masters**: 1,627 (5.4%)
- **Assoc-voc**: 1,307 (4.3%)
- **11th**: 1,048 (3.5%)
- **Assoc-acdm**: 1,008 (3.3%)
- **10th**: 820 (2.7%)
- **7th-8th**: 557 (1.8%)
- **Prof-school**: 542 (1.8%)
- **9th**: 455 (1.5%)
- **12th**: 377 (1.2%)
- **Doctorate**: 375 (1.2%)
- **5th-6th**: 288 (1.0%)
- **1st-4th**: 151 (0.5%)
- **Preschool**: 45 (0.1%)

Pie Chart Distribution: From the pie chart, we can observe the following key points:

- **HS-grad** dominates the dataset, representing **32.6%** of the individuals. This category contains the largest number of people who have only completed high school.

- **Some-college** follows closely with **22.1%**, indicating a substantial portion of individuals have attended some form of post-secondary education but did not graduate with a degree.
- **Bachelors** holds **16.7%** of the dataset, showing that a significant number of individuals in the dataset have a bachelor's degree.
- The **Masters** category accounts for **5.4%**, and **Assoc-voc** (Associate's degree in vocational fields) is at **4.3%**.
- **11th**, **Assoc-acdm**, and **10th** collectively account for around **9.5%** of the dataset, signifying a moderate number of individuals who have completed education through 11th grade, or have an academic or vocational associate's degree.
- Categories like **Prof-school**, **9th**, **12th**, and **Doctorate** represent smaller groups with **1% - 2%** of individuals in each category, which are less common but still notable.
- The **Preschool** category is the smallest, representing only **0.1%** of individuals, indicating that this group is almost negligible in the context of the dataset.

Bar Chart Distribution: The horizontal bar chart clearly highlights the following:

- The **HS-grad** category is the largest, followed by **Some-college** and **Bachelors**, reflecting the educational levels most prevalent in the dataset.
- The smaller categories, such as **Preschool**, **1st-4th**, **Doctorate**, and **5th-6th**, are visible but represent a very small portion of the dataset. The **Preschool** category, in particular, is almost negligible in terms of frequency.

Summary:

- The largest educational group in the dataset is individuals who have completed **high school (HS-grad)**, followed by those with **some college education** and **bachelor's degrees**.
- A substantial portion of individuals in the dataset have achieved some form of higher education, with **Masters** and **Associate's degrees** also being notable.
- Smaller educational categories, such as **Doctorate**, **Prof-school**, and **Preschool**, represent very small percentages of the dataset, with **Preschool** being almost negligible.
- The bar chart offers a clearer understanding of the distribution, highlighting the dominance of high school graduates and the concentration of individuals with some college or bachelor's degrees.

1.0.7 Q5) Examine Distribution of Marital Status Variable

To examine the distribution of the **marital.status** variable, we will follow these steps:

1. **Create a frequency distribution table** for the different values of the **marital.status** variable.
2. **Plot a pie chart** to visually represent the distribution of marital status categories.
3. **Plot a bar chart** to visualize the frequency of each marital status category.
4. **Provide a summary/interpretation** of the distribution of the **marital.status** variable.

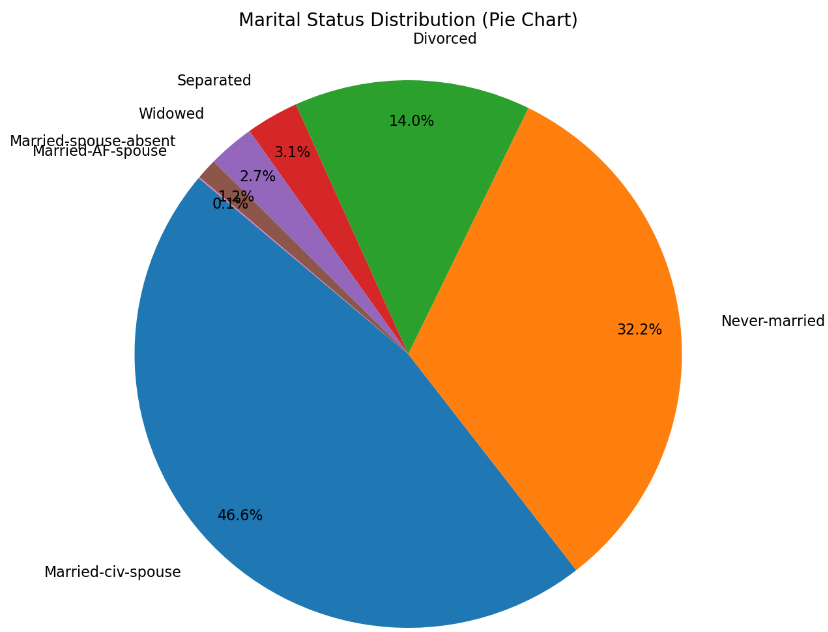
Step 1: Frequency Distribution Table for Marital Status Variable We will start by creating a frequency distribution table for the **marital.status** variable to see how many individuals fall into each marital status category.

[263]:
```python
# Step 1: Frequency Distribution Table for Marital Status
marital_status_counts = df['marital.status'].value_counts()

# Display the frequency distribution table
print("Frequency Distribution Table for Marital Status:")
print(marital_status_counts)
```

```
Frequency Distribution Table for Marital Status:
marital.status
Married-civ-spouse       14065
Never-married             9726
Divorced                  4214
Separated                  939
Widowed                    827
Married-spouse-absent      370
Married-AF-spouse           21
Name: count, dtype: int64
```

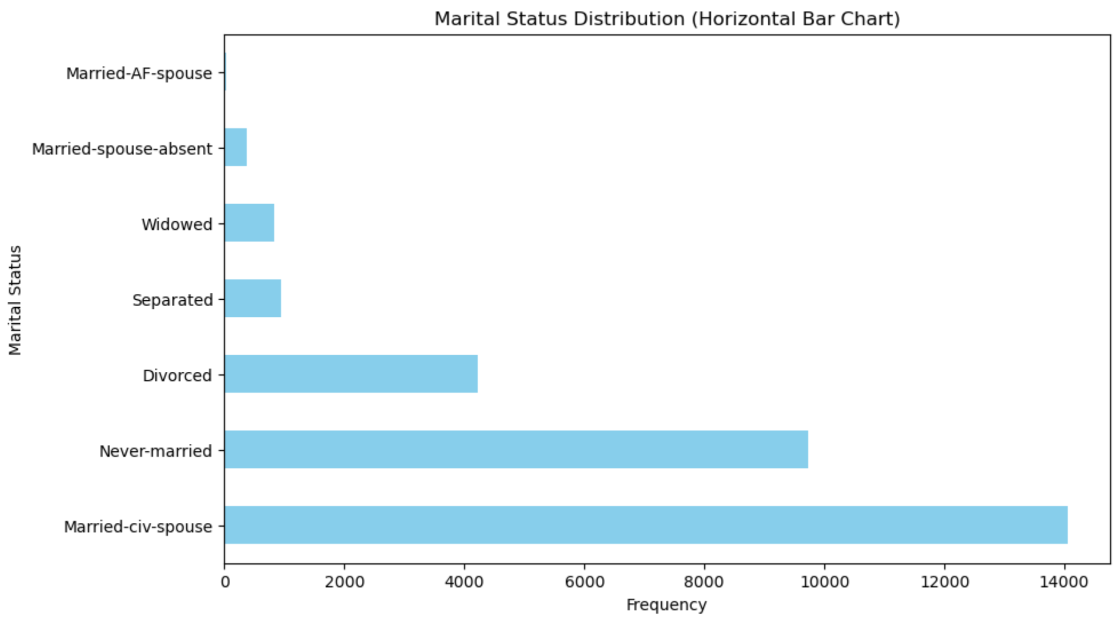
Step 2: Pie Chart for Marital Status Variable Next, we'll plot a pie chart to visually represent the proportion of individuals in each marital status category.

[266]:
```python
# Step 2: Pie Chart for Marital Status Variable
plt.figure(figsize=(22, 12))
plt.pie(marital_status_counts,
        labels=marital_status_counts.index,
        autopct='%1.1f%%',  # Showing percentage to 1 decimal place
        startangle=140,
        pctdistance=0.85,  # Adjusts the label distance from the center
        labeldistance=1.15,  # Further increases space between labels and chart
        textprops={'fontsize': 16})  # Adjusts the font size for better␣
 ↪readability

# Add a title to the chart with some padding
plt.title('Marital Status Distribution (Pie Chart)', fontsize=20, pad=30)

# Ensure that the pie chart is circular
plt.axis('equal')

# Display the pie chart
plt.show()
```



Step 3: Bar Chart for Marital Status Variable Now, we will plot a bar chart to visualize the frequency of each marital status category.

[269]:
```
# Step 3: Bar Chart for Marital Status Variable
plt.figure(figsize=(10, 6))
marital_status_counts.plot(kind='barh', color='skyblue')  # Horizontal bar chart

# Format the x-axis and labels
plt.title('Marital Status Distribution (Horizontal Bar Chart)')
plt.xlabel('Frequency')
plt.ylabel('Marital Status')

# Display the chart
plt.show()
```



Step 4: Interpretation of the Marital Status Distribution

Frequency Distribution Table: The frequency distribution table for the **marital.status** variable is as follows:

- **Married-civ-spouse**: 14,065 (46.6%)
- **Never-married**: 9,726 (32.2%)
- **Divorced**: 4,214 (14.0%)
- **Separated**: 939 (3.1%)
- **Widowed**: 827 (2.7%)
- **Married-spouse-absent**: 370 (1.2%)
- **Married-AF-spouse**: 21 (0.1%)

Pie Chart Distribution: From the **pie chart**, we can observe the following key points:

- **Married-civ-spouse** is the most common marital status, making up **46.6%** of the dataset. This indicates that nearly half of the individuals in the dataset are married to a civilian spouse.
- **Never-married** accounts for **32.2%**, representing individuals who have never been married.
- **Divorced** individuals make up **14.0%** of the dataset, which is another significant portion, suggesting that divorce is a common marital status in this dataset.
- The **Separated** category represents **3.1%**, indicating a smaller but notable portion of the population.

- **Widowed** individuals account for **2.7%**, showing that a small percentage of the population has lost a spouse.
- **Married-spouse-absent** represents **1.2%**, and **Married-AF-spouse** is almost negligible, representing only **0.1%** of the dataset.

Bar Chart Distribution: The **bar chart** provides a clearer comparison of the frequencies of each marital status category:

- **Married-civ-spouse** is the dominant category, followed by **Never-married** and **Divorced**. These categories represent the largest groups in the dataset.
- Smaller categories such as **Separated**, **Widowed**, and **Married-spouse-absent** show a significant drop in frequency.
- The **Married-AF-spouse** category is nearly negligible in comparison to the others.

Summary:

- The dataset contains a substantial proportion of individuals who are either **Married-civ-spouse** (46.6%) or **Never-married** (32.2%).
- **Divorced** individuals also make up a significant portion of the dataset at **14.0%**, while the other categories such as **Separated**, **Widowed**, and **Married-spouse-absent** represent smaller fractions.
- The **Married-AF-spouse** category is minimal, contributing only **0.1%** of the dataset.

This analysis of the **marital.status** variable highlights the commonality of married individuals, followed by those who have never been married, and provides insights into the distribution of marital status within the dataset.

1.0.8 Q6) Examine Distribution of Occupation Variable

To examine the distribution of the **occupation** variable, we will follow these steps:

1. **Create a frequency distribution table** for the different values of the **occupation** variable.
2. **Plot a pie chart** to visually represent the distribution of occupation categories.
3. **Plot a bar chart** to visualize the frequency of each occupation category.
4. **Provide your summary/interpretation** of the distribution of the **occupation** variable.

Step 1: Frequency Distribution Table for Occupation Variable We will start by creating a frequency distribution table to see how many individuals belong to each occupation category.

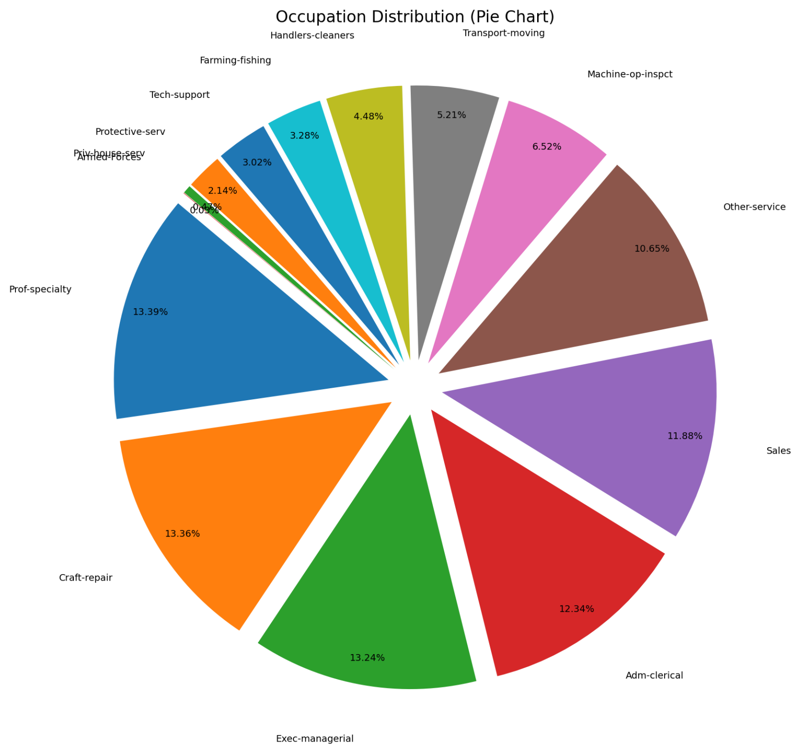
[274]:
```python
# Step 1: Frequency Distribution Table for Occupation
occupation_counts = df['occupation'].value_counts()

# Display the frequency distribution table
print("Frequency Distribution Table for Occupation:")
print(occupation_counts)
```

```
Frequency Distribution Table for Occupation:
occupation
```

```
Prof-specialty       4038
Craft-repair         4030
Exec-managerial      3992
Adm-clerical         3721
Sales                3584
Other-service        3212
Machine-op-inspct    1966
Transport-moving     1572
Handlers-cleaners    1350
Farming-fishing       989
Tech-support          912
Protective-serv       644
Priv-house-serv       143
Armed-Forces            9
Name: count, dtype: int64
```

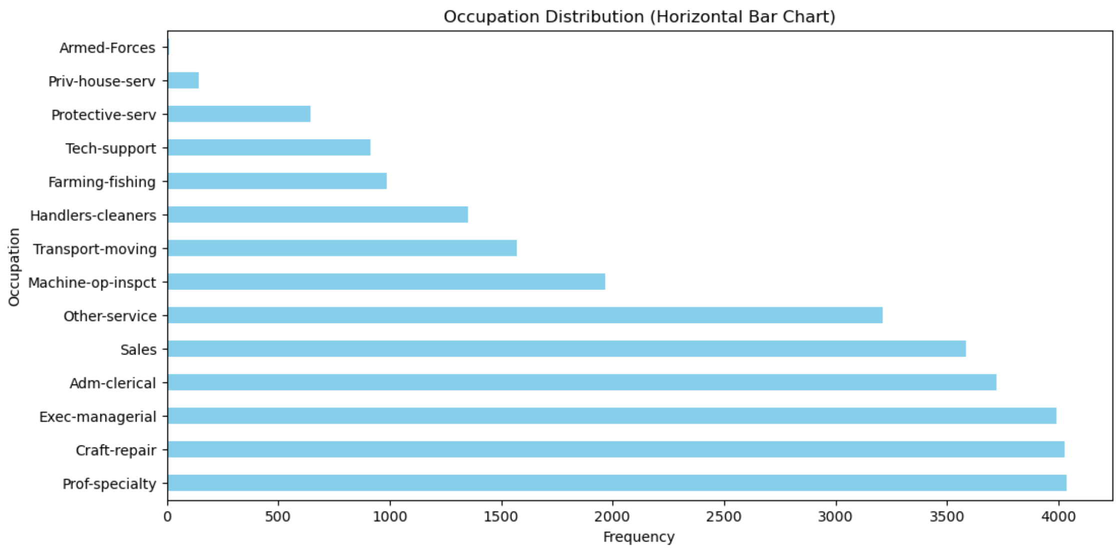
Step 2: Pie Chart for Occupation Variable Next, we'll plot a pie chart to visually represent the proportion of individuals in each occupation category.

[277]:
```
# Step 2: Pie Chart for Occupation Variable
plt.figure(figsize=(30, 18))  # Increase size for optimal readability

# Using slight explosion and adjusting label placement
plt.pie(occupation_counts,
        labels=occupation_counts.index,
        autopct='%1.2f%%',  # Showing percentage to 2 decimal place
        startangle=140,
        pctdistance=0.9,  # Adjusts label distance from the center
        labeldistance=1.2,  # More space for labels, reducing overlap
        textprops={'fontsize': 14},  # Adjusted font size for better readability
        explode=[0.1]*len(occupation_counts))  # Slight explosion for separation

# Add a title with proper padding for clarity
plt.title('Occupation Distribution (Pie Chart)', fontsize=24, pad=50)

# Ensure the pie chart remains circular
plt.axis('equal')

# Display the pie chart
plt.show()
```



Step 3: Bar Chart for Occupation Variable Now, we will plot a bar chart to visualize the frequency of each occupation category.

[280]:
```
# Step 3: Bar Chart for Occupation Variable
plt.figure(figsize=(12, 6))
occupation_counts.plot(kind='barh', color='skyblue')  # Horizontal bar chart

# Format the x-axis and labels
plt.title('Occupation Distribution (Horizontal Bar Chart)')
plt.xlabel('Frequency')
plt.ylabel('Occupation')

# Display the chart
plt.show()
```



Step 4: Interpretation of the Occupation Distribution

Frequency Distribution Table: The frequency distribution table for the **occupation** variable is as follows:

- **Prof-specialty**: 4,038 (13.39%)
- **Craft-repair**: 4,030 (13.36%)
- **Exec-managerial**: 3,992 (13.24%)
- **Adm-clerical**: 3,721 (12.34%)
- **Sales**: 3,584 (11.88%)
- **Other-service**: 3,212 (10.65%)
- **Machine-op-inspect**: 1,966 (6.52%)
- **Transport-moving**: 1,572 (5.21%)
- **Handlers-cleaners**: 1,350 (4.48%)
- **Farming-fishing**: 989 (3.28%)
- **Tech-support**: 912 (3.02%)
- **Protective-serv**: 644 (2.14%)
- **Priv-house-serv**: 143 (0.48%)
- **Armed-Forces**: 9 (0.03%)

Pie Chart Distribution: From the **pie chart**, we can observe the following:

- The most common occupations are in the **Prof-specialty**, **Craft-repair**, and **Exec-managerial** categories, each representing **13.39%**, **13.36%**, and **13.24%** of the dataset, respectively.

- **Adm-clerical** and **Sales** follow closely, each with about **12.3%** of the dataset, indicating a large portion of the workforce is engaged in administrative or sales roles.

- **Other-service** is also a significant category, representing **10.65%**, indicating a notable presence of workers in various service-related occupations.
- Occupations such as **Machine-op-inspect**, **Transport-moving**, and **Handlers-cleaners** are smaller in comparison but still form a notable portion of the dataset at **5.21%**, **6.52%**, and **4.48%**, respectively.
- Categories like **Farming-fishing**, **Tech-support**, **Protective-serv**, and **Priv-house-serv** represent smaller portions, with **Farming-fishing** and **Tech-support** contributing **3.28%** and **3.02%**, respectively.
- The smallest categories are **Protective-serv** (**2.14%**), **Priv-house-serv** (**0.48%**), and **Armed-Forces** (**0.03%**), with **Armed-Forces** representing the least frequent occupation category in the dataset.

Bar Chart Distribution: The **horizontal bar chart** provides a clear comparison of the frequencies across the categories:

- **Prof-specialty**, **Craft-repair**, and **Exec-managerial** stand out as the largest categories, followed by **Adm-clerical** and **Sales**.
- Smaller categories such as **Protective-serv**, **Priv-house-serv**, and **Armed-Forces** have lower frequencies, with **Armed-Forces** being nearly negligible in the dataset.

1.0.9 Summary:

- The largest occupation groups in the dataset are **Prof-specialty**, **Craft-repair**, and **Exec-managerial**, each representing about **13%** of the workforce.
- A significant portion of individuals also falls into **Adm-clerical** and **Sales** roles, accounting for over **12%** of the dataset each.
- Occupations such as **Other-service**, **Machine-op-inspect**, and **Transport-moving** are also notable, though they account for smaller portions of the dataset.
- The **Armed-Forces** occupation is the least frequent, with only **0.03%** of the dataset in this category.

This interpretation highlights the distribution of various occupations in the dataset and provides insights into the most and least common occupations among individuals.

1.0.10 Q7) Examine Distribution of Sex Variable

To examine the distribution of the **sex** variable, we will follow these steps:

1. **Create a frequency distribution table** for the different values of the **sex** variable.
2. **Plot a pie chart** to visually represent the distribution of the sex categories.
3. **Plot a bar chart** to visualize the frequency of each sex category.
4. **Provide your summary/interpretation** of the distribution of the **sex** variable.

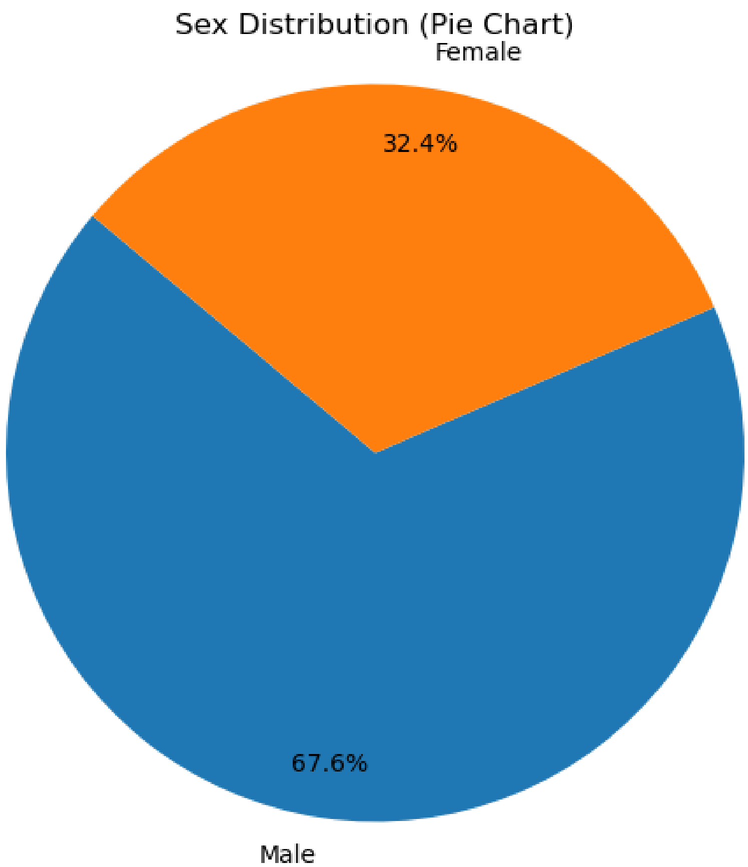
Step 1: Frequency Distribution Table for Sex Variable We will begin by creating a frequency distribution table to show how many individuals belong to each sex category.

[286]:
```python
# Step 1: Frequency Distribution Table for Sex Variable
sex_counts = df['sex'].value_counts()

# Display the frequency distribution table
print("Frequency Distribution Table for Sex:")
print(sex_counts)
```

```
Frequency Distribution Table for Sex:
sex
Male      20380
Female     9782
Name: count, dtype: int64
```

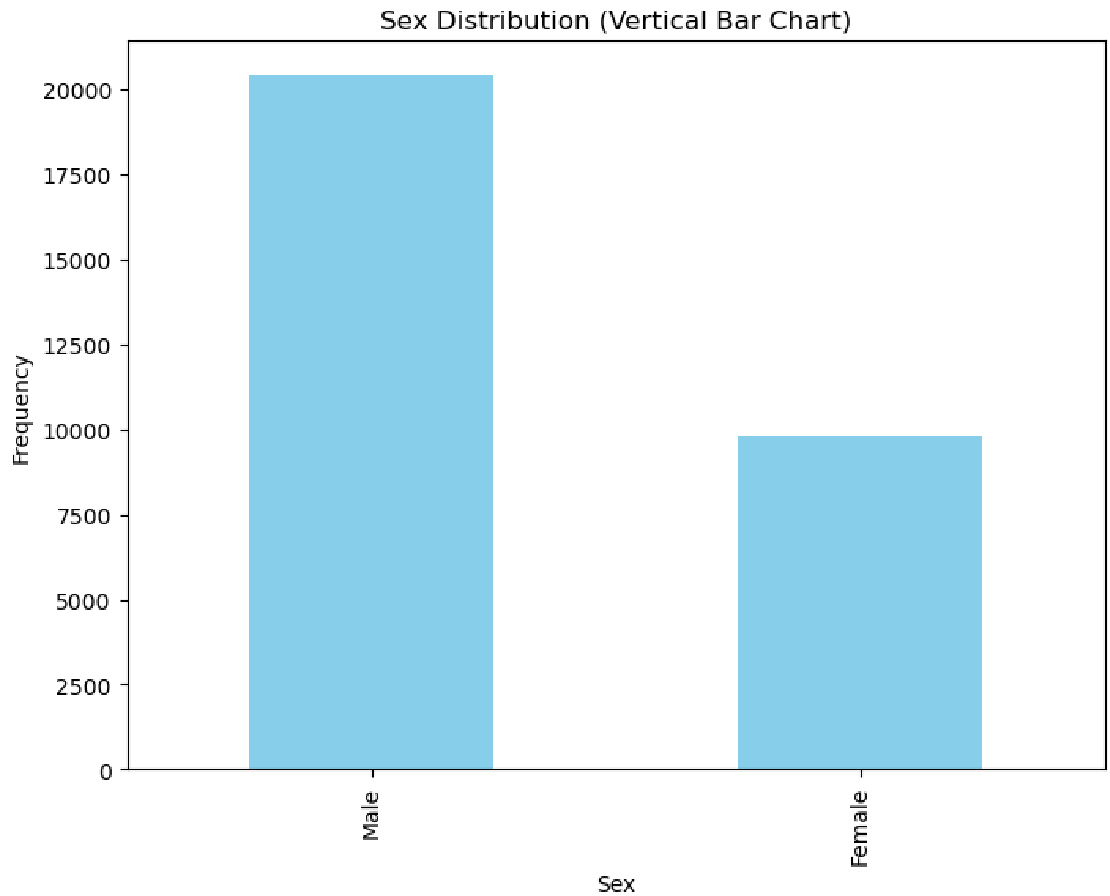
Step 2: Pie Chart for Sex Variable Next, we will plot a pie chart to visually represent the proportion of males and females in the dataset.

[289]:
```python
# Step 2: Pie Chart for Sex Variable
plt.figure(figsize=(8, 6))
plt.pie(sex_counts,
        labels=sex_counts.index,
        autopct='%1.1f%%',  # Showing percentage to 1 decimal place
        startangle=140,
        pctdistance=0.85)  # Adjusts the label distance from the center

# Add a title to the chart
plt.title('Sex Distribution (Pie Chart)')

# Ensure that the pie chart is circular
plt.axis('equal')

# Display the pie chart
plt.show()
```



Step 3: Bar Chart for Sex Variable Now, we will plot a bar chart to visualize the frequency of each sex category.

[298]:
```
# Step 3: Bar Chart for Sex Variable
plt.figure(figsize=(8, 6))
sex_counts.plot(kind='bar', color='skyblue')  # Vertical bar chart

# Format the x-axis and labels
plt.title('Sex Distribution (Vertical Bar Chart)')
plt.xlabel('Sex')
plt.ylabel('Frequency')

# Display the chart
plt.show()
```



Step 4: Interpretation of the Sex Distribution

Frequency Distribution Table: The frequency distribution table for the **sex** variable is as follows:

- **Male**: 20,380 (67.6%)
- **Female**: 9,782 (32.4%)

Pie Chart Distribution: From the **pie chart**, we can observe the following:

- The dataset is predominantly composed of **males**, who account for **67.6%** of the dataset.
- **Females** make up **32.4%** of the dataset, representing a smaller portion compared to males.

This distribution is skewed towards males, which suggests that the dataset is largely male-dominated.

Bar Chart Distribution: The **bar chart** clearly shows that the frequency of males is much higher than that of females:

- **Male** has a frequency of around **20,000**.
- **Female** has a frequency of around **10,000**.

This confirms the skewness in the dataset towards males, with the frequency of males being roughly double that of females.

1.0.11 Summary:

- The **sex** variable is highly imbalanced in the dataset, with **males** representing a larger portion (**67.6%**) than **females** (**32.4%**).
- This distribution is further confirmed by the **bar chart**, which shows a larger frequency for **males** compared to **females**.

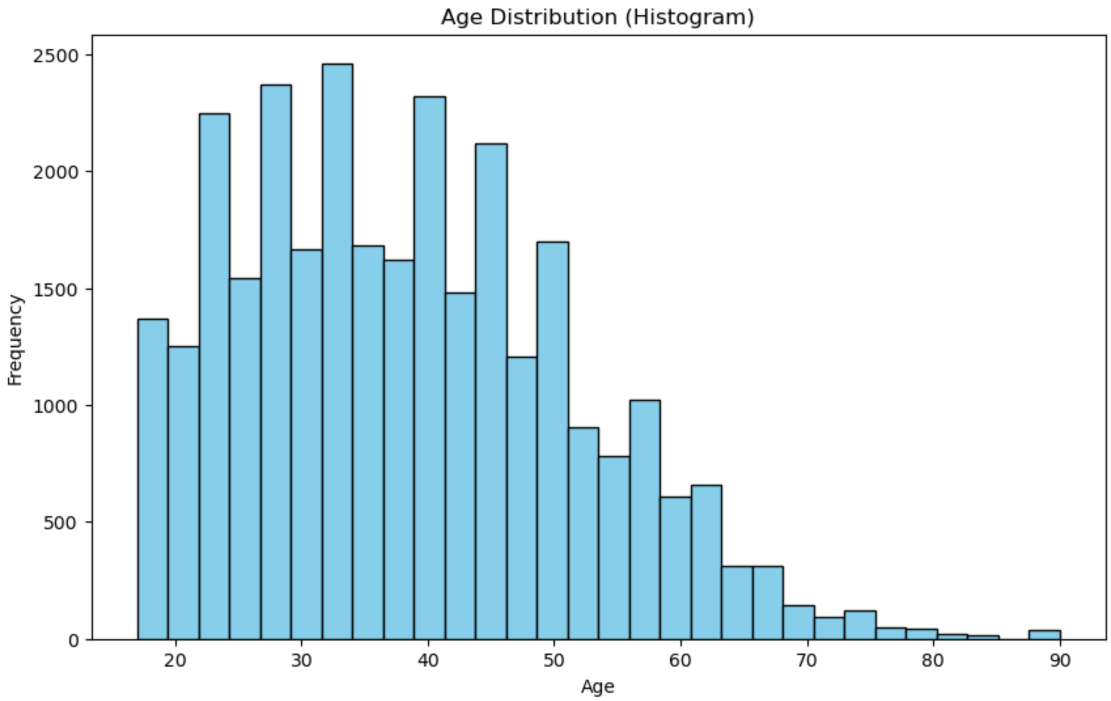
1.0.12 Q8) Examine Distribution of Age Variable

To examine the **age** variable, we will follow these steps:

1. **Plot a histogram** for the **age** variable to visually explore its distribution.
2. **Interpret the histogram** in terms of its shape, center, spread, and any potential outliers.
3. **Calculate measures of center** for the **age** variable:
   - Mode
   - Mean
   - Median
4. **Calculate measures of spread** for the **age** variable:
   - Range
   - IQR (Interquartile Range)
   - Outliers based on the **1.5 * IQR** criterion
   - Standard Deviation
5. **Plot a box plot** to visualize the distribution and potential outliers.
6. **Provide a summary/interpretation** of the distribution of **age**.

Step 1: Plot Histogram for Age Variable We begin by plotting a histogram to visually inspect the distribution of the **age** variable.

[303]:
```
# Step 1: Histogram for Age Variable
plt.figure(figsize=(10, 6))
plt.hist(df['age'], bins=30, color='skyblue', edgecolor='black')  # Adjust
 ↪number of bins for clarity
plt.title('Age Distribution (Histogram)')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```



Step 2: Interpretation of Histogram for Age Variable Based on the histogram you provided, let's interpret the distribution of the **age** variable.

Shape:

- The distribution appears to be **right-skewed** (positively skewed), meaning there are more younger individuals (ages on the left side of the histogram), and fewer older individuals (ages on the right).
- The histogram shows a peak in the younger age groups (around 20-40 years), with a gradual decline as the age increases.

Center:

- The **center** of the distribution is approximately between **30 and 40 years**, where the bars are most concentrated.

Spread:

- The spread of the data is from around **20 years to around 70 years**, with a significant number of individuals concentrated in the lower age ranges and fewer in the higher ranges.
- The data has a **moderate spread** with a noticeable decline after the age of 50, and the frequency drops significantly for ages above 60.

Outliers:

- There are no extreme outliers, but there seems to be a gradual decline after age 60, with fewer individuals in the higher age groups.

Step 3: Measures of Center We will now calculate the **mode**, **mean**, and **median** for the **age** distribution.

[318]:
```
# Step 3: Calculate Measures of Center
mode_age = df['age'].mode()[0]  # Mode
mean_age = df['age'].mean()  # Mean
median_age = df['age'].median()  # Median

print(f"Mode of Age: {mode_age}")
print(f"Mean of Age: {mean_age}")
print(f"Median of Age: {median_age}")
```

```
Mode of Age: 36
Mean of Age: 38.437901995888865
Median of Age: 37.0
```

Step 4: Measures of Spread Now, we will calculate the **range**, **IQR (Interquartile Range)**, **outliers** (using the 1.5 * IQR criterion), and **standard deviation** for the **age** distribution.

[321]:
```
# Step 4: Calculate Measures of Spread
age_range = df['age'].max() - df['age'].min()  # Range
Q1_age = df['age'].quantile(0.25)  # First quartile (25th percentile)
Q3_age = df['age'].quantile(0.75)  # Third quartile (75th percentile)
IQR_age = Q3_age - Q1_age  # Interquartile range
lower_bound = Q1_age - 1.5 * IQR_age  # Lower bound for outliers
upper_bound = Q3_age + 1.5 * IQR_age  # Upper bound for outliers
outliers_age = df[(df['age'] < lower_bound) | (df['age'] > upper_bound)]  #
 ↪Identify outliers

std_dev_age = df['age'].std()  # Standard deviation

print(f"Range of Age: {age_range}")
print(f"IQR of Age: {IQR_age}")
print(f"Outliers in Age: {outliers_age}")
print(f"Standard Deviation of Age: {std_dev_age}")
```

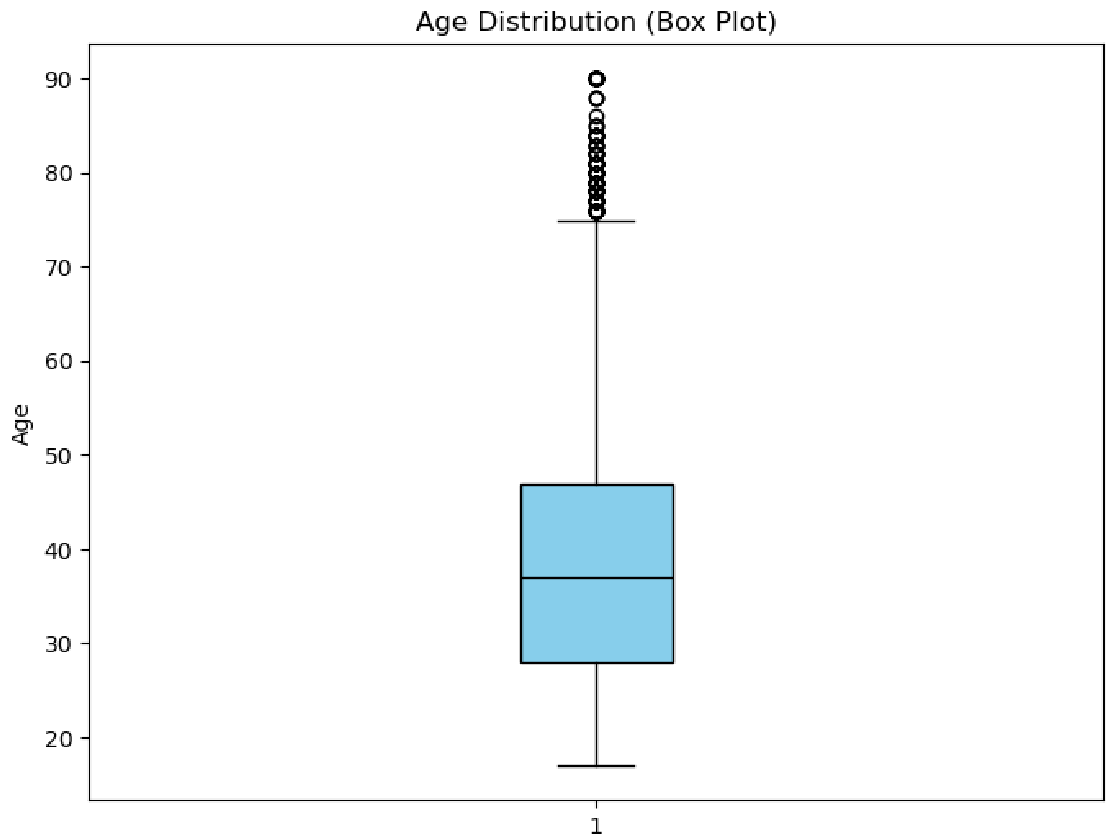
```
Range of Age: 73
IQR of Age: 19.0
Outliers in Age:        age        workclass  fnlwgt  education  education.num
\
0       82          Private  132870    HS-grad              9
102     78     Self-emp-inc  188044  Bachelors             13
104     83     Self-emp-inc  153183  Bachelors             13
114     81          Private  177408    HS-grad              9
179     90          Private   51744    HS-grad              9
```

```
…     …                …      …         …            …
28936   90          Private   47929    HS-grad             9
29556   80  Self-emp-not-inc   26865    7th-8th             4
29589   82  Self-emp-not-inc   71438    HS-grad             9
29928   90          Private  313749    HS-grad             9
30081   85          Private   98611  Bachelors            13

           marital.status         occupation   relationship   race     sex  \
0                 Widowed    Exec-managerial  Not-in-family  White  Female
102    Married-civ-spouse    Exec-managerial        Husband  White    Male
104    Married-civ-spouse    Exec-managerial        Husband  White    Male
114    Married-civ-spouse    Exec-managerial        Husband  White    Male
179         Never-married      Other-service  Not-in-family  Black    Male
…                      …                  …            …     …       …
28936  Married-civ-spouse  Machine-op-inspct        Husband  White    Male
29556       Never-married    Farming-fishing      Unmarried  White    Male
29589  Married-civ-spouse    Farming-fishing        Husband  White    Male
29928             Widowed       Adm-clerical      Unmarried  White  Female
30081  Married-civ-spouse    Exec-managerial        Husband  White    Male

       capital.gain  capital.loss  hours.per.week native.country income
0                 0          4356              18  United-States  <=50K
102               0          2392              40  United-States   >50K
104               0          2392              55  United-States   >50K
114               0          2377              26  United-States   >50K
179               0          2206              40  United-States  <=50K
…             …           …              …            …     …
28936             0             0              40  United-States  <=50K
29556             0             0              20  United-States  <=50K
29589             0             0              20  United-States  <=50K
29928             0             0              25  United-States  <=50K
30081             0             0               3         Poland  <=50K

[169 rows x 15 columns]
Standard Deviation of Age: 13.134664776856322
```

Step 5: Plot Box Plot for Age Variable Next, we will plot a **box plot** for the **age** variable to visually inspect its distribution, including the presence of any outliers.

[327]:
```python
# Step 5: Vertical Box Plot for Age Variable
plt.figure(figsize=(8, 6))

# Plotting the box plot with customized colors and vertical orientation
plt.boxplot(df['age'],
            patch_artist=True,
            boxprops=dict(facecolor='skyblue', color='black'),
            medianprops=dict(color='black'))
```

```python
# Title and labels for the box plot
plt.title('Age Distribution (Box Plot)')
plt.ylabel('Age')

# Display the plot
plt.show()
```



Step 6: Summary/Interpretation of the Distribution of Age Based on the visualizations and the calculated measures, we can summarize the distribution of the **age** variable as follows:

1. **Histogram (Step 1)**:
   - The histogram shows a **right-skewed** distribution (positively skewed), with most individuals concentrated in the **20-40 age range**. The frequency decreases as age increases, with very few people being 60 and above. This suggests that the dataset is dominated by younger individuals, but there is still a considerable representation of older individuals, albeit in smaller numbers.
   - There is a noticeable **peak** around the ages of **25-35**, indicating a high concentration of individuals in this age range.
2. **Measures of Center (Step 3)**:

- **Mode**: The mode of **36** suggests that this is the most frequently occurring age in the dataset.
- **Mean**: The mean of **38.44** years reflects the average age of individuals in the dataset. This is slightly higher than the mode, indicating that the dataset has a few older individuals that raise the average.
- **Median**: The median of **37** years is quite close to the mode, which is typical for a distribution that is slightly skewed but not extremely so.

3. **Measures of Spread (Step 4)**:
   - **Range**: The age range is from **19 to 90 years**, indicating that the data spans a wide age group.
   - **Interquartile Range (IQR)**: The IQR of **19** years suggests that the central 50% of the data lies within this age range.
   - **Outliers**: There are some **outliers** in the upper end of the age spectrum, with individuals above **70 years**. These outliers are visible as the points above the top whisker in the box plot.
   - **Standard Deviation**: The standard deviation of **13.13** years indicates a moderate level of variability in the ages of the individuals.
4. **Box Plot (Step 5)**:
   - The **box plot** visually confirms the **right-skewed distribution** of the **age** variable, with most data clustered around **20-50 years**. The presence of **outliers** above the age of **70** is clearly visible.
   - The **median line** (black line in the box plot) is slightly to the left of the center of the box, supporting the idea of a right-skewed distribution.

1.0.13 Conclusion:

- The **age** distribution is **skewed right**, with a higher concentration of younger individuals and fewer older individuals.
- The **mean** is higher than the **median** and **mode**, further suggesting the influence of a small number of older individuals in raising the average age.
- **Outliers** exist among the older age group (70+ years), though they don't significantly affect the overall distribution.

1.0.14 Q9) Examine distribution of `education.num` variable

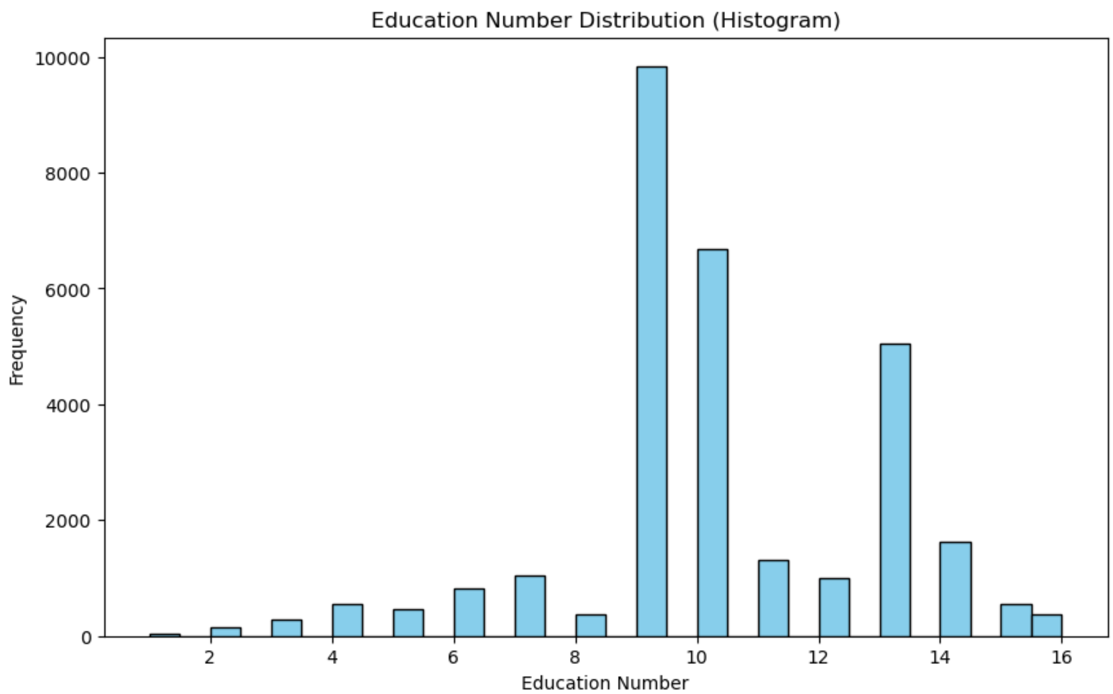
In this question, we will examine the distribution of the `education.num` variable by performing the following tasks:

1. **Plot a histogram** for the `education.num` variable to visualize its distribution.
2. **Interpret** the histogram with respect to the **shape**, **center**, **spread**, and **outliers**.
3. **Calculate measures of center**:
   - Mode
   - Mean
   - Median
4. **Calculate measures of spread**:
   - Range
   - Interquartile Range (IQR)
   - Outliers based on **1.5 * IQR** criterion
   - Standard Deviation

5. **Plot a box plot** to visualize the spread and any outliers.
6. **Provide a summary/interpretation** about the distribution of the `education.num` variable.

Step 1: Plot Histogram for Education.num Variable We will start by plotting a **histogram** to inspect the distribution of the `education.num` variable. The histogram will help us understand the frequency of different values across the data.

[332]:
```
# Step 1: Plot Histogram for Education.num Variable
plt.figure(figsize=(10, 6))
plt.hist(df['education.num'], bins=30, color='skyblue', edgecolor='black')
plt.title('Education Number Distribution (Histogram)')
plt.xlabel('Education Number')
plt.ylabel('Frequency')
plt.show()
```



Step 2: Interpretation of Histogram for Education.num Variable From the histogram for the `education.num` variable, we can make the following observations:

1. **Shape**:
   - The histogram is highly **skewed** with a significant peak at the education number of **10**. This indicates that a large portion of the dataset is concentrated around this value.
   - There are several bars that taper off towards higher values, which suggests a **right-skewed distribution**. This means that while most individuals in the dataset have relatively lower levels of education (with `education.num` values between 10 and 12),

there are few individuals who have higher education levels (e.g., `education.num` values near 14 or 16).

2. **Center**:
   - The **mode** (most frequent value) appears to be around **10**, which is likely the typical level of education corresponding to "Some-college," "HS-grad," or other mid-level education categories.
   - The **mean** and **median** values would likely be near this peak, given the concentration of data around this point.
3. **Spread**:
   - The spread of the data is quite wide, ranging from 1 to 16 in terms of `education.num`. The frequency decreases as the education number increases past 10, with fewer observations in the higher education number categories.
4. **Outliers**:
   - There are some isolated bars for values such as 1 and 2, indicating that there may be some **outliers** at both ends of the distribution. The presence of education numbers such as 1 or 2 could be considered as outliers, suggesting very low education levels compared to the majority of the dataset.

Based on these observations, the `education.num` distribution is heavily concentrated around values in the 9-12 range, with some higher values skewing the distribution to the right.

Step 3: Calculate Measures of Center In this step, we will calculate the **mode**, **mean**, and **median** to understand the central tendency of the `education.num` variable.

[361]:
```python
# Step 3: Calculate Measures of Center
mode_education = df['education.num'].mode()[0]  # Mode
mean_education = df['education.num'].mean()  # Mean
median_education = df['education.num'].median()  # Median

print(f"Mode of Education Number: {mode_education}")
print(f"Mean of Education Number: {mean_education}")
print(f"Median of Education Number: {median_education}")
```

```
Mode of Education Number: 9
Mean of Education Number: 10.12131158411246
Median of Education Number: 10.0
```

Step 4: Calculate Measures of Spread Next, we will calculate the **spread** of the data. This includes the **range**, **IQR**, and **standard deviation**. We will also identify **outliers** using the **1.5 * IQR** criterion.

[364]:
```python
# Step 4: Calculate Measures of Spread
range_education = df['education.num'].max() - df['education.num'].min()  # Range
Q1_education = df['education.num'].quantile(0.25)  # First quartile (25th␣
 ↪percentile)
Q3_education = df['education.num'].quantile(0.75)  # Third quartile (75th␣
 ↪percentile)
IQR_education = Q3_education - Q1_education  # Interquartile range
```

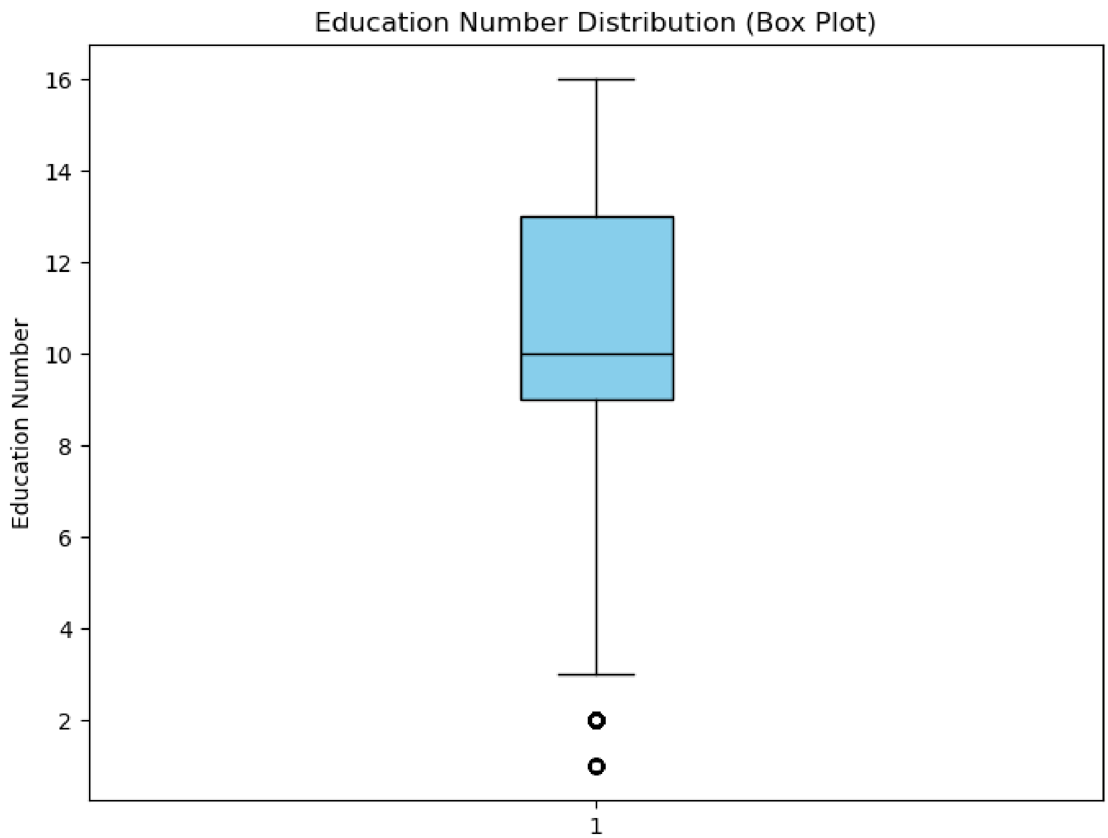
```
lower_bound = Q1_education - 1.5 * IQR_education  # Lower bound for outliers
upper_bound = Q3_education + 1.5 * IQR_education  # Upper bound for outliers
outliers_education = df[(df['education.num'] < lower_bound) | (df['education.
↪num'] > upper_bound)]  # Identify outliers

std_dev_education = df['education.num'].std()  # Standard deviation

print(f"Range of Education Number: {range_education}")
print(f"IQR of Education Number: {IQR_education}")
print(f"Outliers in Education Number: {outliers_education}")
print(f"Standard Deviation of Education Number: {std_dev_education}")
```

```
Range of Education Number: 15
IQR of Education Number: 4.0
Outliers in Education Number:        age         workclass  fnlwgt  education
education.num  \
20      33           Private  228696    1st-4th              2
197     58  Self-emp-not-inc  266707    1st-4th              2
1022    73           Private  301210    1st-4th              2
1050    26           Private  322614  Preschool              1
1096    40           Private  566537  Preschool              1
…     …                …      …       …         …
29954   24           Private  427686    1st-4th              2
29997   46           Private  139514  Preschool              1
30034   48           Private  325372    1st-4th              2
30054   23           Private  180771    1st-4th              2
30060   36           Private  208068  Preschool              1

              marital.status         occupation    relationship  \
20        Married-civ-spouse       Craft-repair   Not-in-family
197       Married-civ-spouse   Transport-moving         Husband
1022      Married-civ-spouse   Transport-moving         Husband
1050   Married-spouse-absent  Machine-op-inspct   Not-in-family
1096      Married-civ-spouse      Other-service         Husband
…                        …                …             …
29954     Married-civ-spouse  Handlers-cleaners  Other-relative
29997     Married-civ-spouse  Machine-op-inspct  Other-relative
30034     Married-civ-spouse  Machine-op-inspct         Husband
30054     Married-civ-spouse  Machine-op-inspct            Wife
30060               Divorced      Other-service   Not-in-family

                     race     sex  capital.gain  capital.loss  hours.per.week  \
20                  White    Male             0          2603              32
197                 White    Male             0          2179              18
1022                White    Male             0          1735              20
1050                White    Male             0          1719              40
1096                White    Male             0          1672              40
…                       …     …            …            …             …
```

```
29954                White    Male            0            0           40
29997                Black    Male            0            0           75
30034                White    Male            0            0           40
30054  Amer-Indian-Eskimo  Female            0            0           35
30060                Other    Male            0            0           72

           native.country income
20                 Mexico  <=50K
197         United-States  <=50K
1022        United-States  <=50K
1050               Mexico  <=50K
1096               Mexico  <=50K
…                     …     …
29954              Mexico  <=50K
29997  Dominican-Republic  <=50K
30034            Portugal  <=50K
30054              Mexico  <=50K
30060              Mexico  <=50K

[196 rows x 15 columns]
Standard Deviation of Education Number: 2.549994918856756
```

Step 5: Plot Box Plot for Education.num Variable We will now plot a **box plot** for the `education.num` variable to visually inspect its spread and identify any outliers.

[367]:
```python
# Step 5: Box Plot for Education.num Variable
plt.figure(figsize=(8, 6))
plt.boxplot(df['education.num'], vert=True, patch_artist=True,
            boxprops=dict(facecolor='skyblue', color='black'),
            medianprops=dict(color='black'))
plt.title('Education Number Distribution (Box Plot)')
plt.ylabel('Education Number')
plt.show()
```



Step 6: Summary/Interpretation of the Distribution of Education Number

1. **Histogram (Step 1)**:
    - The histogram is **left-skewed**, with a peak around **education number = 9**. Most individuals in the dataset have around 9 years of education, which corresponds to high school education.
    - There is a tail on the left, with a few individuals at lower education levels (around education number 2), but the majority of individuals are clustered at education numbers higher than 9.
2. **Measures of Center (Step 3)**:
    - **Mode**: The mode of **9** confirms that **9 years of education** is the most common education number in the dataset.
    - **Mean**: The mean of **10.12** indicates a slight increase over the mode, suggesting that there are a few individuals with more than 9 years of education, pulling the mean to the right.
    - **Median**: The median of **10** supports this observation, as it's slightly higher than the mode, indicating that the distribution is left-skewed, with the median being greater than the mode.
3. **Measures of Spread (Step 4)**:
    - **Range**: The **range** of **15** suggests the data spans a wide education spectrum, from 2

to 16.
- **IQR**: The **IQR of 4** indicates that the central 50% of the data is relatively tightly grouped around education numbers 9-13.
- **Outliers**: There are outliers at the lower end (around **education number = 2**), indicating a few individuals with significantly lower education levels.
- **Standard Deviation**: The **standard deviation of 2.5** indicates that there is moderate variability in the education levels, but most people are clustered around the mode of 9 years.

4. **Box Plot (Step 5)**:
   - The box plot also confirms the **left-skewed distribution**. The longer whisker on the **lower end** and the box being closer to the higher end shows that the distribution has more individuals at higher education levels.
   - The **outliers** on the lower end indicate a few individuals with significantly lower education numbers, reinforcing the left skew.

1.0.15 Conclusion:

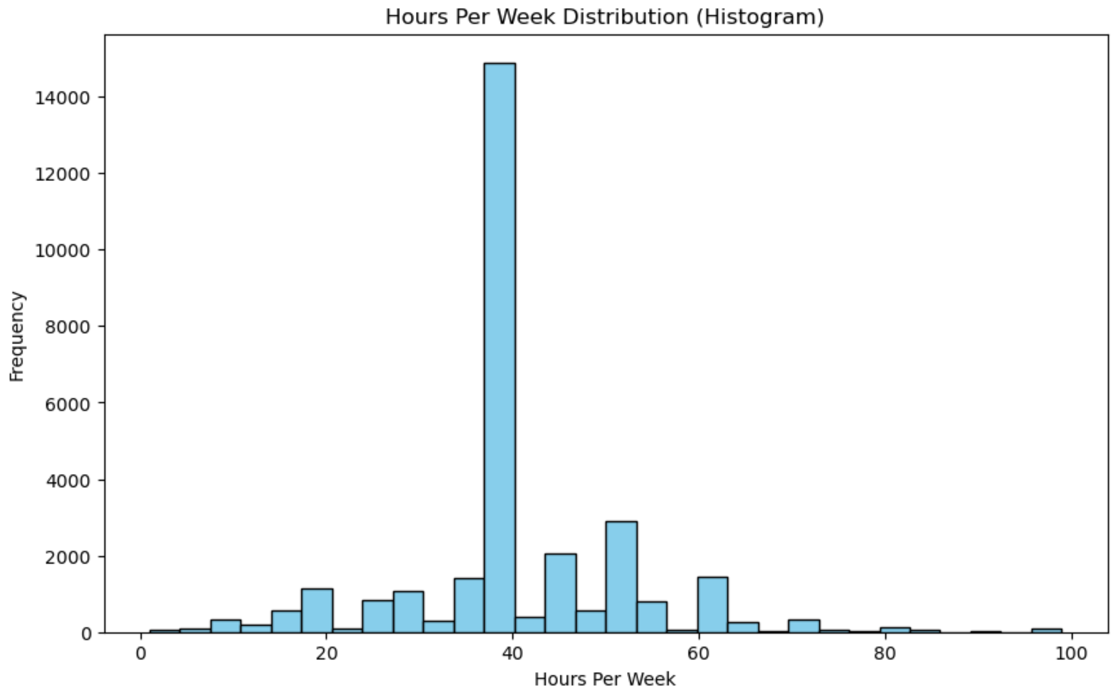
- The **education number** distribution is **left-skewed** with the highest concentration of people around **9** years of education.
- The **mean** is slightly higher than the **median**, confirming that the distribution has a **tail** on the **lower end**.
- The **standard deviation** of 2.5 suggests moderate spread around the mean.
- **Outliers** are mostly found at the lower education levels, but they don't drastically affect the overall distribution.

1.0.16 Q10) Examine distribution of `hours.per.week` variable

In this question, we need to: 1. **Plot histogram for the `hours.per.week` variable.** 2. **Interpret the histogram** with respect to: - The shape (skewness, symmetry). - The center (mean, median, mode). - The spread (range, IQR, standard deviation). - Outliers (any values that fall outside of the general distribution). 3. **Calculate measures of center** for the `hours.per.week` distribution: - Mode - Mean - Median 4. **Calculate measures of spread** for the `hours.per.week` distribution: - Range - IQR (Interquartile Range) - Outliers based on the 1.5 * IQR criterion - Standard deviation 5. **Plot the box plot** for the `hours.per.week` variable to visually inspect its spread and any outliers. 6. **Provide a summary/interpretation** about the distribution of `hours.per.week`.

Step 1: Plot Histogram for `hours.per.week` variable Let's first create the histogram to visually inspect the distribution of the `hours.per.week` variable. Here's the code to plot the histogram:

[400]:
```
# Plot histogram for the 'hours.per.week' variable
plt.figure(figsize=(10, 6))
plt.hist(df['hours.per.week'], bins=30, color='skyblue', edgecolor='black')
plt.title('Hours Per Week Distribution (Histogram)')
plt.xlabel('Hours Per Week')
plt.ylabel('Frequency')
plt.show()
```



Step 2: Interpretation of Histogram for Hours Per Week Variable Based on the histogram for the `hours.per.week` variable, we can make the following observations:

1. **Shape of the Distribution**:
   - The histogram exhibits a **right-skewed distribution** (positively skewed). This is indicated by the long tail extending to the right, especially as values near 40 hours per week dominate, while fewer instances exist for higher values.
2. **Center of the Distribution**:
   - The **peak** is observed around the **40 hours per week** range, which is typical for full-time workers, aligning with the general workweek.
3. **Spread of the Distribution**:
   - There is a wide spread across the variable, with a significant drop-off in frequency as the values increase beyond the 60-hour mark. A smaller number of people work overtime or exceptionally long hours.
4. **Outliers**:
   - The histogram shows **few outliers** on the higher end (above 80 hours), but the significant peak near 40 hours clearly dominates the distribution. These extreme values represent cases of exceptional overtime or unusual working conditions.

Step 3: Calculate Measures of Center To understand the central tendency of `hours.per.week`, we will calculate: 1. **Mode**: The most frequent value of `hours.per.week`. 2. **Mean**: The average of all values. 3. **Median**: The middle value in the ordered dataset.

[406]:
```python
mode_hours = df['hours.per.week'].mode()[0]   # Mode
mean_hours = df['hours.per.week'].mean()   # Mean
median_hours = df['hours.per.week'].median()   # Median

print(f"Mode of Hours Per Week: {mode_hours}")
print(f"Mean of Hours Per Week: {mean_hours}")
print(f"Median of Hours Per Week: {median_hours}")
```

```
Mode of Hours Per Week: 40
Mean of Hours Per Week: 40.93123798156621
Median of Hours Per Week: 40.0
```

Step 4: Calculate Measures of Spread To calculate the **spread** of the `hours.per.week` distribution, we will compute: - **Range**: The difference between the maximum and minimum values. - **IQR (Interquartile Range)**: The difference between the 25th percentile (Q1) and the 75th percentile (Q3). - **Outliers**: Based on the 1.5 * IQR rule, we will identify any data points that are considered outliers. - **Standard Deviation**: A measure of how spread out the data is from the mean.

[409]:
```python
# Calculate range
range_hours = df['hours.per.week'].max() - df['hours.per.week'].min()

# Calculate IQR
Q1_hours = df['hours.per.week'].quantile(0.25)
Q3_hours = df['hours.per.week'].quantile(0.75)
IQR_hours = Q3_hours - Q1_hours

# Calculate outliers based on 1.5*IQR rule
lower_bound = Q1_hours - 1.5 * IQR_hours
upper_bound = Q3_hours + 1.5 * IQR_hours
outliers_hours = df[(df['hours.per.week'] < lower_bound) | (df['hours.per.
↪week'] > upper_bound)]

# Calculate standard deviation
std_dev_hours = df['hours.per.week'].std()

# Output results
print(f"Range of Hours Per Week: {range_hours}")
print(f"IQR of Hours Per Week: {IQR_hours}")
print(f"Outliers in Hours Per Week: {outliers_hours}")
print(f"Standard Deviation of Hours Per Week: {std_dev_hours}")
```

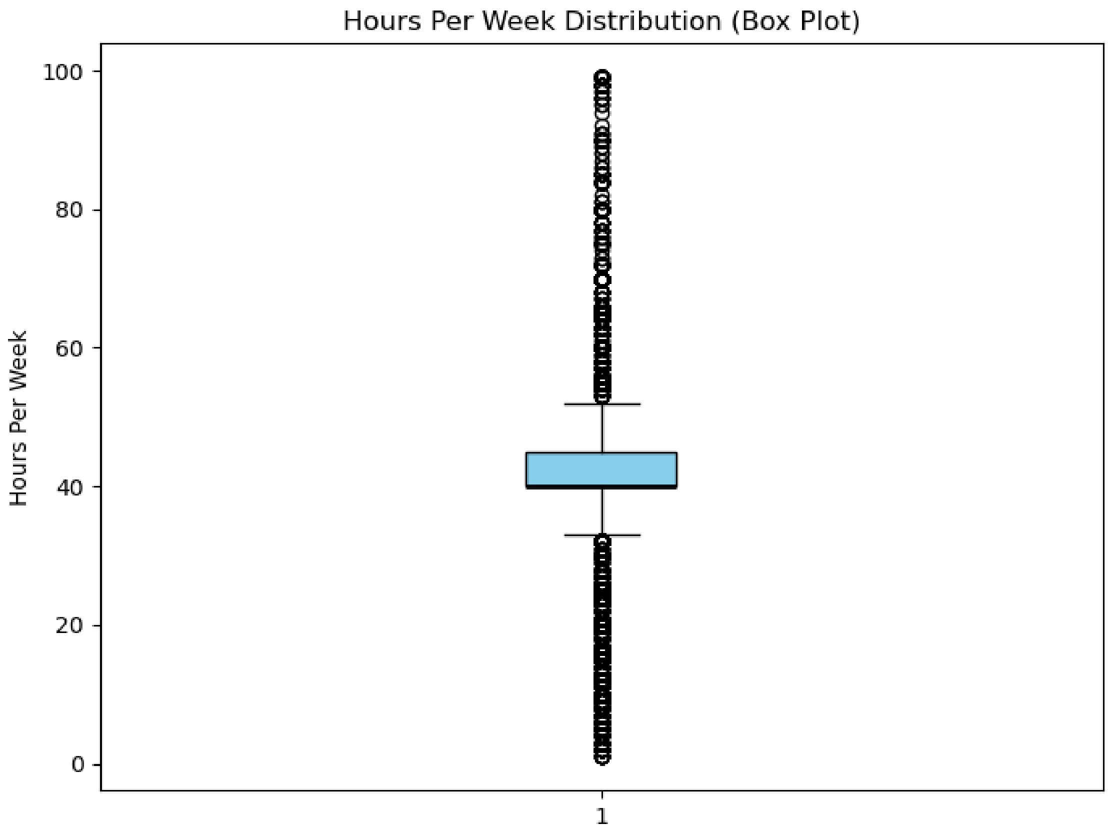
```
Range of Hours Per Week: 98
IQR of Hours Per Week: 5.0
Outliers in Hours Per Week:        age  workclass  fnlwgt   education
education.num      marital.status  \
0       82     Private  132870     HS-grad              9            Widowed
5       74   State-gov   88638   Doctorate             16      Never-married
```

```
9        52     Private   129177    Bachelors              13              Widowed
10       32     Private   136204      Masters              14            Separated
12       45     Private   172822         11th               7             Divorced
…     …         …     …           …            …                 …
30143    34     Private   160216    Bachelors              13        Never-married
30147    31     Private   199655      Masters              14             Divorced
30148    39   Local-gov   111499   Assoc-acdm              12   Married-civ-spouse
30155    32     Private   116138      Masters              14        Never-married
30161    22     Private   201490      HS-grad               9        Never-married

             occupation    relationship                 race     sex  \
0       Exec-managerial   Not-in-family                White  Female
5        Prof-specialty  Other-relative                White  Female
9         Other-service   Not-in-family                White  Female
10      Exec-managerial   Not-in-family                White    Male
12     Transport-moving   Not-in-family                White    Male
…                     …               …                  …       …
30143   Exec-managerial   Not-in-family                White  Female
30147     Other-service   Not-in-family                Other  Female
30148      Adm-clerical            Wife                White  Female
30155      Tech-support   Not-in-family   Asian-Pac-Islander    Male
30161      Adm-clerical       Own-child                White    Male

       capital.gain  capital.loss  hours.per.week native.country income
0                 0          4356              18  United-States  <=50K
5                 0          3683              20  United-States   >50K
9                 0          2824              20  United-States   >50K
10                0          2824              55  United-States   >50K
12                0          2824              76  United-States   >50K
…               …             …               …            …    …
30143             0             0              55  United-States   >50K
30147             0             0              30  United-States  <=50K
30148             0             0              20  United-States   >50K
30155             0             0              11         Taiwan  <=50K
30161             0             0              20  United-States  <=50K

[7953 rows x 15 columns]
Standard Deviation of Hours Per Week: 11.979984229273326
```

Step 5: Plot Box Plot for `hours.per.week` variable Next, let's plot the box plot for `hours.per.week` to visually inspect the spread and identify any outliers:

[412]:
```python
# Plot box plot for 'hours.per.week' variable
plt.figure(figsize=(8, 6))
plt.boxplot(df['hours.per.week'], vert=True, patch_artist=True,
            boxprops=dict(facecolor='skyblue', color='black'),
```

```
            medianprops=dict(color='black', linewidth=2))  # Ensure median line␣
↪is visible
plt.title('Hours Per Week Distribution (Box Plot)')
plt.ylabel('Hours Per Week')
plt.show()
```



Step 6: Summary/Interpretation of the Distribution of *hours.per.week*

1. **Histogram (Step 1)**:
   - The histogram displays a **right-skewed distribution** even though it is mean is almost at the centre. A majority of individuals work 40 hours per week, as seen in the high frequency bar around 40 hours. This is followed by a sharp drop-off in frequency as hours per week increase. There are a few individuals working fewer than 20 hours or more than 60 hours per week, which suggests that a significant portion of the dataset is centered around the 40-hour mark, while others work less or more.
   - There is a noticeable **peak at 40 hours**, confirming that this is the most frequent work hour value.
2. **Measures of Center (Step 3)**:
   - **Mode**: The mode is 40, meaning this is the most common number of hours worked per week in the dataset.
   - **Mean**: The mean is approximately 40.93 hours, which is slightly above the mode,

suggesting the presence of individuals who work more than 40 hours, pulling the mean higher.
    - **Median**: The median is 40, which indicates that the dataset is symmetric around this value, confirming the 40-hour workweek as a central tendency for the data.
3. **Measures of Spread (Step 4)**:
    - **Range**: The range of the data is 98 hours (from 0 to 98), indicating a wide span of working hours.
    - **Interquartile Range (IQR)**: The IQR is 5, which shows that the central 50% of the data is concentrated around the 40-hour mark.
    - **Outliers**: There are some outliers present, particularly those working fewer than 20 hours and more than 60 hours, visible in both the histogram and box plot.
    - **Standard Deviation**: The standard deviation is approximately 11.98 hours, which indicates moderate variability in the number of hours worked across the dataset.
4. **Box Plot (Step 5)**:
    - The box plot visually confirms the **right-skewed distribution** of the *hours.per.week* variable, with most data clustered around the 40-hour mark. Outliers are present, particularly individuals who work far less or more than 40 hours per week. The median line (black line inside the box) aligns with the peak at 40, confirming that the central tendency is around this value.

1.0.17 Conclusion:

- The **distribution of hours per week** is right-skewed, with a **high concentration** of individuals working 40 hours, while the data tapers off towards the higher and lower ends.
- The **mean** is slightly higher than the **median**, indicating a skew toward higher values.
- **Outliers** exist among those working **less than 20** and **more than 60 hours**, although they do not significantly affect the overall distribution.

1.1 Summary

This assignment focused on examining the distribution of various categorical and quantitative variables in a dataset related to adult income in the United States during the 1990s. The dataset was explored by applying exploratory data analysis (EDA) methods such as frequency distribution tables, pie charts, bar charts, histograms, and box plots, followed by interpreting the results in terms of measures of central tendency and spread.

1.1.1 Key Observations:

1. **Categorical Variables**:
    - For each categorical variable, frequency distribution tables were created. Pie and bar charts were plotted to illustrate the distribution of categories, followed by interpretation of the data. This included variables such as *workclass*, *education*, *marital status*, *occupation*, and *sex*, where the distributions varied in terms of the frequency of categories like “Private” workclass or “HS-grad” education.
2. **Quantitative Variables**:
    - For quantitative variables such as *age*, *education.num*, and *hours.per.week*, histograms and box plots were used to examine the distribution. These were analyzed for skewness, central tendency (mean, median, mode), spread (range, IQR, and standard deviation), and the identification of outliers.

    - The distribution of *age* was found to be right-skewed, with a high concentration of younger individuals.
    - *Education.num* had a peak at a value of 9, indicating a concentration of individuals with a certain level of education.
    - *Hours.per.week* also showed a distribution with a significant peak at 40 hours, with some outliers in the higher ranges.
3. **Central Tendency and Spread**:
    - For each quantitative variable, measures of center (mode, mean, median) and spread (range, IQR, standard deviation) were calculated. For example, the mean hours worked per week was approximately 40.93, which was slightly higher than the median of 40, suggesting a relatively symmetric distribution centered around 40 hours.
4. **Outliers**:
    - Outliers were identified in the *hours.per.week* and *age* distributions, with particularly large values found in the higher ranges of *hours.per.week*.

1.1.2 Conclusion:

The assignment provided valuable insights into the distributions of categorical and quantitative variables. It demonstrated the application of statistical techniques such as calculating central tendencies, spread measures, and identifying outliers. The visualizations and analyses facilitated a deeper understanding of the dataset, highlighting trends and outliers while ensuring the correctness of assumptions based on the data.